Operating System Quiz 30-04-2026

Memory Management

Hi, GAURANG. When you submit this form, the owner will see your name and email address.

* Required

1. Name: *

Gaurang Tyagi

2. Class Roll: *

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3. Which of the following is NOT a memory management technique?

- ○ Paging
- ○ Segmentation
- ◉ Compilation
- ○ Swapping

4. In paging, the logical address is divided into:

- ○ Segment number and offset
- ◉ Page number and offset
- ○ Frame number and offset
- ○ Base and limit

5. What is the main advantage of paging?

- ○ Eliminates internal fragmentation

- (selected) Eliminates external fragmentation
- Reduces page faults
- Avoids context switching

6. Internal fragmentation occurs in:

- (selected) Paging
- Segmentation
- Both
- Neither

7. External fragmentation occurs in:

- Paging
- (selected) Segmentation
- Both
- Neither

8. Page size is typically:

- Variable
- (selected) Fixed
- Increasing dynamically
- Dependent on program

9. In segmentation, each segment has:

- Page number and frame
- (selected) Base and limit
- Offset and frame

- ○ Index and block

10. Which register holds the starting address of a segment?

- ○ Limit register
- ● Base register
- ○ Program counter
- ○ Stack pointer

11. The Translation Lookaside Buffer (TLB) is used to:

- ○ Store pages
- ● Speed up address translation
- ○ Replace pages
- ○ Store instructions

12. Effective memory access time is minimized when:

- ○ TLB hit ratio is low
- ● TLB hit ratio is high
- ○ Page size is large
- ○ Frame size is small

13. Which of the following suffers from external fragmentation?

- ○ Paging
- ● Segmentation
- ○ Demand paging
- ○ Virtual memory

14. In paging, page table stores:

- ○ Logical addresses
- ◉ Frame Number
- ○ Segment limits
- ○ Cache entries

15. A page fault occurs when:

- ◉ Page is not in memory
- ○ Page size is too large
- ○ TLB is full
- ○ CPU is idle

16. Demand paging loads:

- ○ Entire process
- ◉ Only required pages
- ○ All segments
- ○ Only stack

17. What is thrashing?

- ○ Excess CPU usage
- ◉ Excess paging activity
- ○ Memory leak
- ○ Disk failure

18. Which is used to reduce external fragmentation in segmentation?

- [x] Compaction
- [ ] Swapping
- [ ] Paging
- [ ] Caching

19. Logical address in segmentation is:

- [ ] (Page, Offset)
- [x] (Segment, Offset)
- [ ] (Frame, Offset)
- [ ] (Index, Block)

20. If page size = 1 KB and logical address space = 16 bits, number of pages =

- [ ] 16
- [ ] 32
- [x] 64
- [ ] 256

21. Which technique combines paging and segmentation?

- [ ] Virtual memory
- [x] Segmented paging
- [ ] Demand paging
- [ ] Swapping

22. TLB is:

- [x] Cache memory
- [ ] Secondary memory

- ○ Register
- ○ Main memory

23. What happens on a TLB miss?

- ○ Process terminates
- ○ Page fault occurs
- ◉ Page table is accessed
- ○ CPU halts

24. Fragmentation inside a page is called:

- ○ External fragmentation
- ◉ Internal fragmentation
- ○ Compaction
- ○ Paging fault

25. If a segment limit is exceeded, it results in:

- ○ Page fault
- ◉ Trap/exception
- ○ Segmentation fault
- ○ Interrupt

26. Which one provides better logical view to programmer?

- ○ Paging
- ◉ Segmentation
- ○ Swapping

- ◯ Virtual memory

27. In multi-level paging, the advantage is:

- ◯ Faster access
- ◉ Reduced page table size
- ◯ No page faults
- ◯ No fragmentation

28. Which of the following increases page fault rate?

- ◯ Large memory
- ◯ Locality of reference
- ◉ Thrashing
- ◯ TLB

29. Segmentation is visible to:

- ◯ Programmer
- ◯ Hardware only
- ◯ OS only
- ◯ Compiler only